

Vikas Singh

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EXPERIENCE

Associate Consultant – Data Engineering

Dec 2024 – Present

Hakkoda (IBM)

- Led migration of enterprise healthcare platforms from Amazon Redshift and MySQL to Snowflake, converting 50,000+ SQL queries and 8,000+ database objects; reduced critical batch execution time from 9 to 2.5 hours (72%).
- Designed and built the Snowflake Importer, Aggregator, Distributor and Exporter Frameworks, deploying, testing, and validating all across Test/PP1/PP2 environments for multiple healthcare tenants.
- Built a reusable Data Validation & QA framework that reduced validation time from 4 hours per table to 8–9 minutes for 200+ tables, automating row-hash, schema, statistical, and duplicate checks across Snowflake, MySQL, Redshift, PostgreSQL, SQLite, CSV, and Excel.
- Engineered a configurable Patch Installer & Deployment Automation platform with full Snowflake support, reducing manual deployment effort by 30+ engineering hours/week and improving release efficiency by ~40%.
- Enhanced six Java Spring Boot middleware modules integrated with Snowflake; owned API validation, integration testing, and regression testing across five environments from Local through Production.
- Led root-cause analysis for critical production issues across Snowflake, MySQL, S3, middleware. Implemented AWS CloudWatch monitoring and drove fixes with cross-functional and client stakeholders.
- Mentored 8–9 engineers on Snowflake best practices, reducing dependency on senior team members and accelerating project delivery.

PROJECTS

HR Chatbot - Agentic RAG | *LangGraph, FastAPI, Pinecone, LLM's, Tools*

[Link](#)

- Designed and built an end-to-end Agentic RAG application: an India-centric HR assistant orchestrated by a LangGraph state machine that routes, retrieves, grades, and rewrites queries across a private knowledge base and live web search.
- Built an Agentic RAG retrieval pipeline using vector embeddings and similarity search across PDF, DOCX, MD, and TXT documents; implemented an 8-node routing, retrieval, grading, rewriting, generation, and web-search fallback workflow.
- Shipped a full-stack production application with FastAPI endpoints (/chat, /ingest, /kb/stats), admin ingestion console, chat UI with citations/trace view, and DuckDB audit logging.

DataValidator Pro | *Streamlit, Python, Multi-DB*

[Link](#)

- Independently extended a data validation tool originally built for a Snowflake migration into a standalone, multi-database framework, generalizing it to support Snowflake, MySQL, Redshift, PostgreSQL, SQLite, CSV, and Excel sources.
- Implemented row-hash, schema, statistical, and duplicate validation with configurable validation workflows and automated Excel/CSV reconciliation reporting.
- Built Streamlit/CLI-based validation workflows with cell-level table-to-table comparison and color-coded discrepancy reporting, enabling rapid identification and analysis of data mismatches across heterogeneous sources.

TECHNICAL SKILLS

Data Engineering: Python, SQL, Snowflake, Amazon Redshift, ETL/ELT, Data Migration, Data Validation, Data Modeling, SQL Optimization, Apache Spark, Apache Airflow

AI / GenAI: Langchain, LangGraph, RAG, MCP, LLM Orchestration, FastAPI, Vector Search, Vector DB's, Embeddings, Prompt Engineering , Tools / Function Calling, AI Agents, Streamlit

Cloud: AWS (S3, CloudWatch, DynamoDB, Secrets Manager, Lambda Functions), Docker, CI/CD

CERTIFICATIONS

SnowPro Core Certified — Snowflake

Databricks Certified Associate Developer for Apache Spark – Python — Databricks

EDUCATION

B.Tech in Computer Science (Artificial Intelligence)

2021 – 2025

Swami Keshvanand Institute of Technology